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import torch
from transformers import BertTokenizer, BertForSequenceClassification
from transformers import Trainer, TrainingArguments
from datasets import Dataset
import pandas as pd

# ♦ Load your IMDB dataset
df = pd.read_csv(
    "/content/IMDB Dataset.csv",
    engine="python",
    on_bad_lines="skip"
)
# Convert labels
df['label'] = df['sentiment'].map({'positive': 1, 'negative': 0})
df = df[['review', 'label']]

# Use small subset (for demo)
df = df.sample(2000, random_state=42)

dataset = Dataset.from_pandas(df)

# ♦ Tokenizer
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

def tokenize(example):
    return tokenizer(example['review'], truncation=True, padding='max_length')

dataset = dataset.map(tokenize, batched=True)

dataset = dataset.train_test_split(test_size=0.2)

# ♦ Model
model = BertForSequenceClassification.from_pretrained("bert-base-uncased", num_labels=2)

# ♦ Training args
training_args = TrainingArguments(
    output_dir="./results",
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    logging_dir="./logs",
)

# ♦ Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=dataset['train'],
```

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    eval_dataset=dataset['test'],
)

# ♦ Train
trainer.train()

# ♦ Test prediction
inputs = tokenizer("This movie was amazing!", return_tensors="pt")
outputs = model(**inputs)

pred = torch.argmax(outputs.logits, dim=1).item()

print("Prediction:", "Positive " if pred == 1 else "Negative ")

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/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning: The secret `HF_TOKEN` does not exist in your Colab secrets. To authenticate with the Hugging Face Hub, create a token in your settings tab. You will be able to reuse this secret in all of your notebooks. Please note that authentication is recommended but still optional to access public models.
  warnings.warn(

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tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]

```

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vocab.txt: 0.00B [00:00, ?B/s]

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Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable (see https://huggingface.co/docs/huggingface_hub/en/quick-start#authentication).
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a token as an environment variable (see https://huggingface.co/docs/huggingface_hub/en/quick-start#authentication).

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tokenizer.json: 0.00B [00:00, ?B/s]

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Map: 0%|          | 0/2000 [00:00<?, ? examples/s]

```

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config.json: 0%|          | 0.00/570 [00:00<?, ?B/s]

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model.safetensors: 0%|          | 0.00/440M [00:00<?, ?B/s]

```

Loading weights: 0%| | 0/199 [00:00<?, ?it/s]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased

Key	Status
cls.predictions.transform.dense.bias	UNEXPECTED
cls.seq_relationship.weight	UNEXPECTED
cls.predictions.transform.LayerNorm.weight	UNEXPECTED
cls.predictions.transform.LayerNorm.bias	UNEXPECTED
cls.seq_relationship.bias	UNEXPECTED
cls.predictions.transform.dense.weight	UNEXPECTED
cls.predictions.bias	UNEXPECTED
classifier.weight	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture
 - MISSING :those params were newly initialized because missing from the checkpoint
- `logging\_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD\_LOGGING\_DIR`
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: DataLoader object will be initialized with a new DataLoader object.
super().__init__(loader)

<div>

<progress value='30' max='400' style='width:300px; height:20px; vertical-align:middle;'>
[30/400 05:18 < 1:10:07, 0.09 it/s, Epoch 0.14/2]

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Step Training Loss